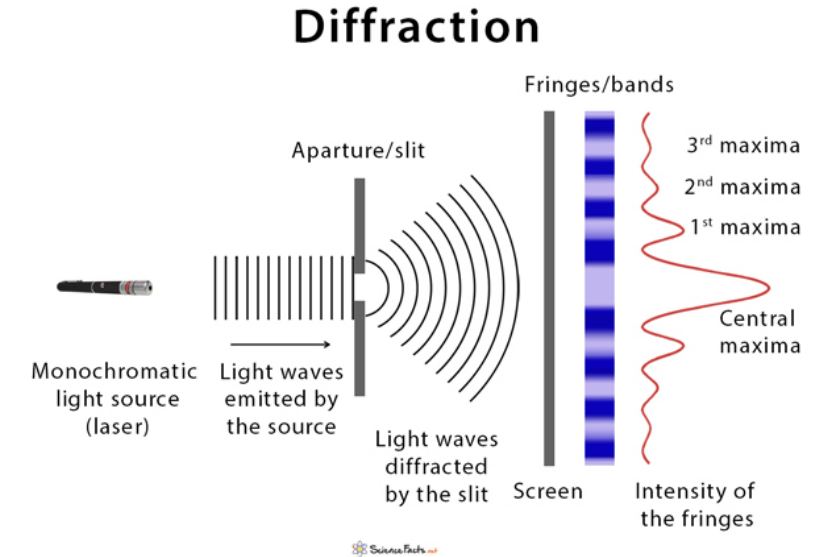
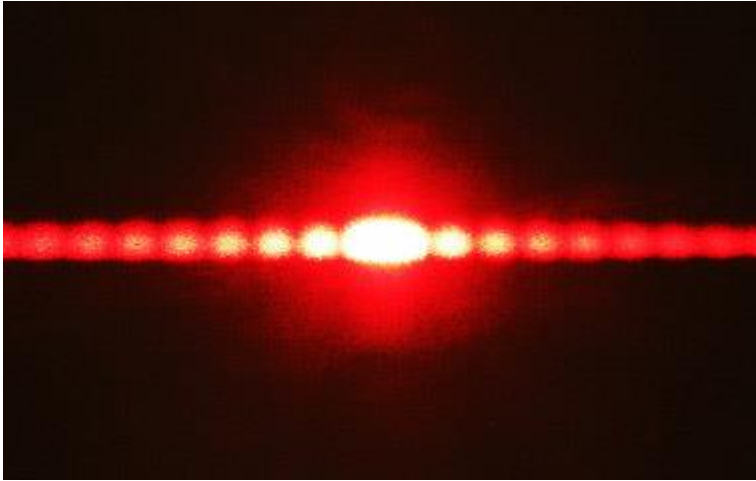


Wave and Optics

Part-6

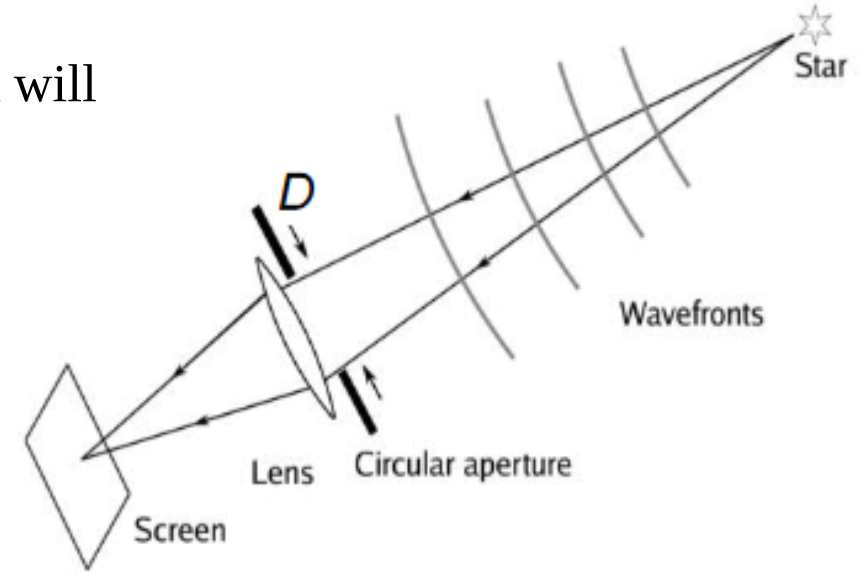
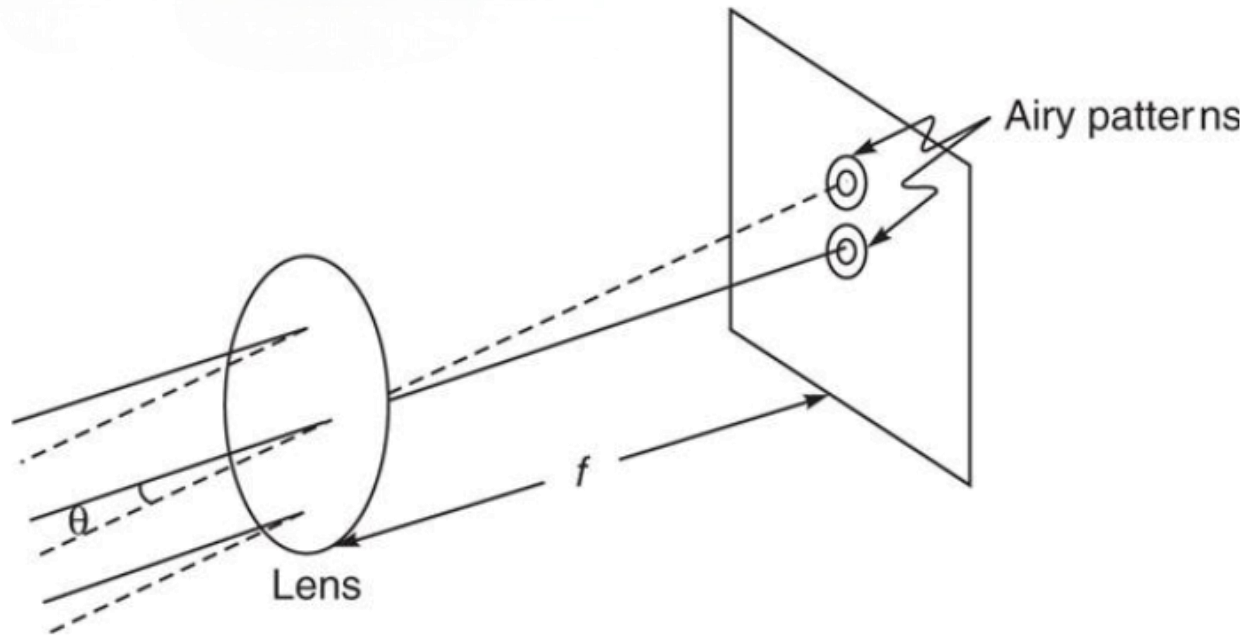


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Limit of resolution

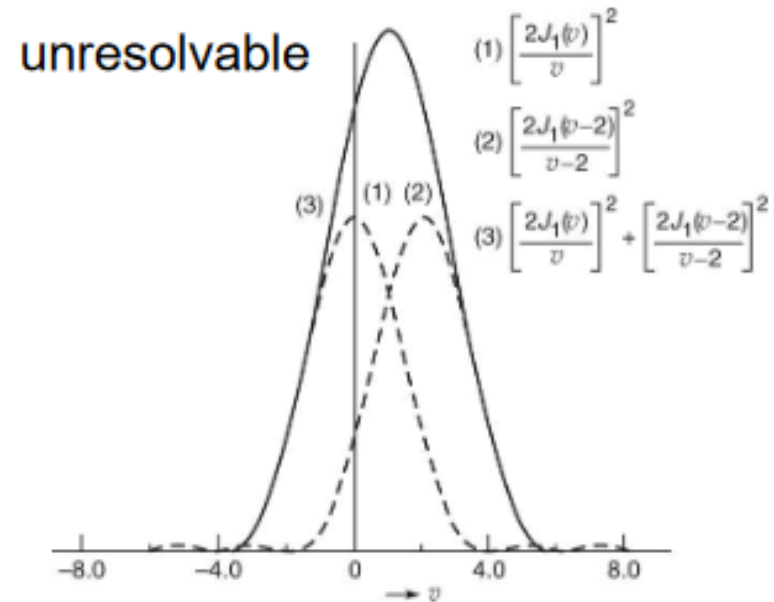
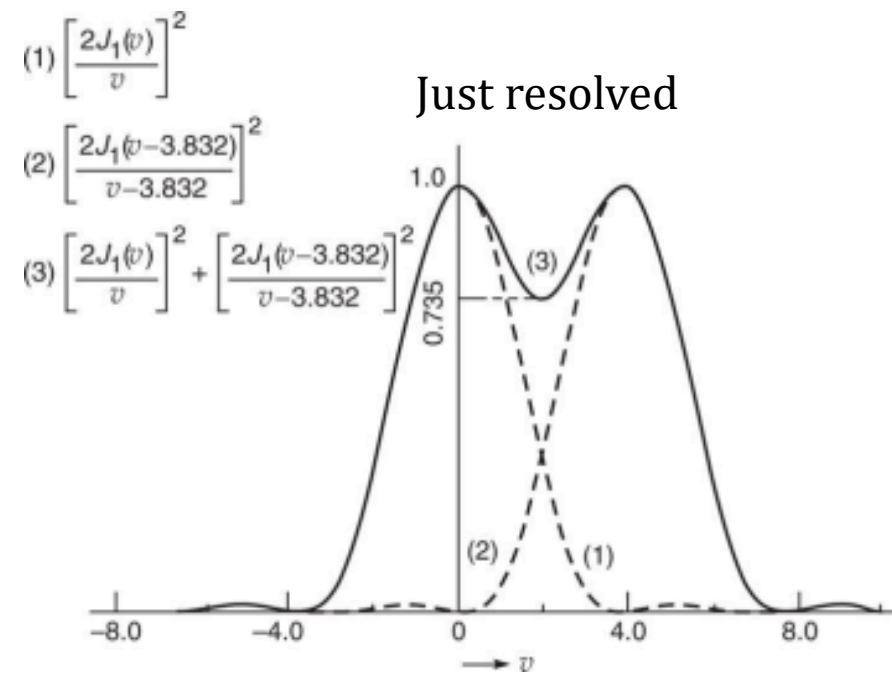
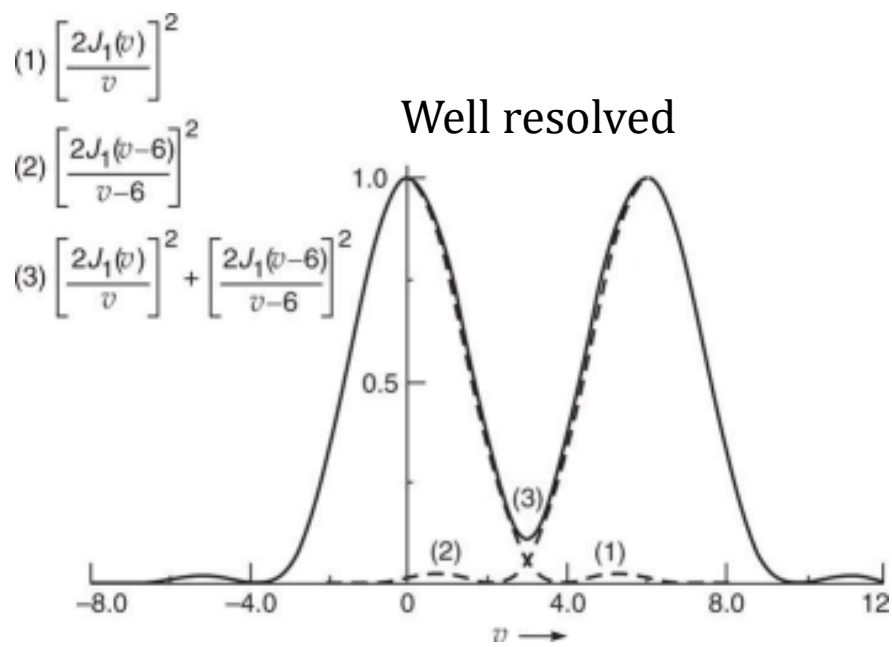
- Let us consider resolving two stars by a telescope
- At the focus each of these point sources (being so far away) from earth will form its own Airy pattern
- Diameter of the Airy disc will depend on λ_0 , f , and diameter D of the objective lens.



Radius of the 1st dark ring, which contains 84% of the diffracted energy is: $\approx 1.22 \lambda/D$

⇒ Larger the value of aperture, smaller will be the spread of the diffracted light

⇒ For higher resolution, one chooses a telescope of larger objective diameter



Resolving power: The resolving power R characterizes the minimum separation $\delta\lambda$ of the central wavelengths of two closely spaced equally intense lines, which are considered to be just resolved. R can as well be expressed in terms of wave number or frequency, and it is defined as

$$\mathcal{R} = \frac{\lambda}{\delta\lambda}$$

Read: Text book Optics by Ajoy Ghatak

Example: Assuming resolving power of human eye is determined by diffraction effects alone, calculate the maximum distance at which two approaching motorcycles separated by a distance of 2 m can be resolved by the eye. (assume $\lambda = 6 \times 10^{-5} \text{ cm}$)

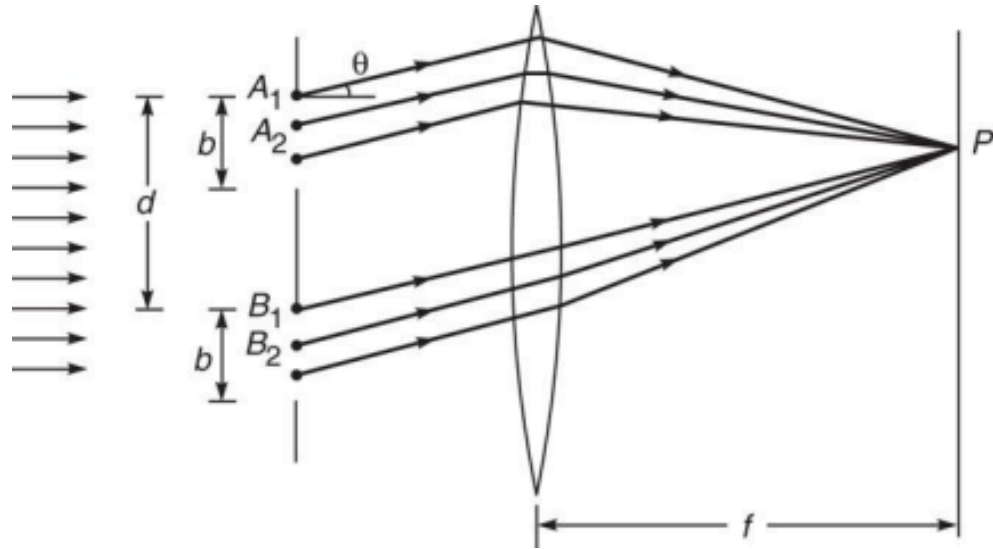
Let the distance be $l \Rightarrow \underbrace{\Delta\theta}_{\substack{\downarrow \\ \text{Diffraction divergence}}} \times l = 1.22 \frac{\lambda}{D} \times l = 2 \text{ m}$

$$\Rightarrow l = \frac{2 (m) \times D}{1.22 \times \lambda}$$

$$\Rightarrow l = \frac{2 \times 2 \times 10^{-3}}{1.22 \times 6 \times 10^{-7}} \frac{m^2}{m}$$

$$\Rightarrow l \approx 5.5 \text{ km}$$

Two slit diffraction pattern:



$$\text{At P, } E = \underbrace{E_1 + E_2}_{\text{Electric fields of the diffracted lights from apertures A and B respectively}}$$

Electric fields of the diffracted lights from apertures A and B respectively

Electric fields of the diffracted light from a single aperture

$$E = A \frac{\sin \beta}{\beta} \cos(\omega t - \beta)$$

- pairs of points: $(A_1, B_1), (A_2, B_2), \dots$, two points in each pair is separated by d

$\Rightarrow$ optical path difference between lights reaching P in one pair will be given by $d \sin \theta$

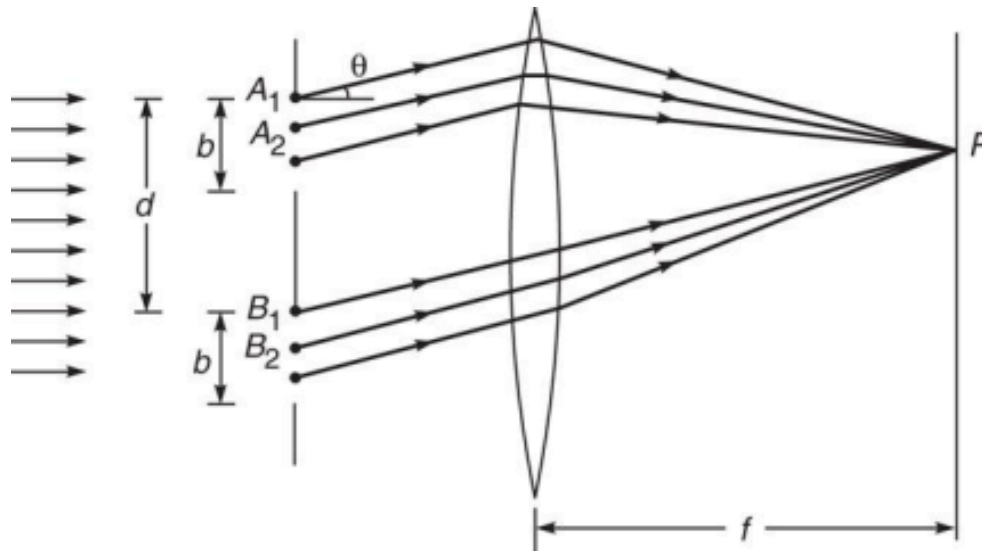
$\Rightarrow$ A phase difference of $\Phi_1 = \frac{2\pi}{\lambda} d \sin \theta$ will accrue between them.

Thus net electric field at P :

$$E = A \frac{\sin \beta}{\beta} \left[\underbrace{\cos(\omega t - \beta)}_{E_1} + \underbrace{\cos(\omega t - \beta - \Phi_1)}_{E_2} \right]$$

$$\beta = \frac{\pi}{\lambda} b \sin \theta \text{ (see Slide 8, Lecture-8)}$$

$$\Phi_1 = \frac{2\pi}{\lambda} d \sin \theta$$



Using

$$\cos A + \cos B = 2 \cos \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B)$$

We get

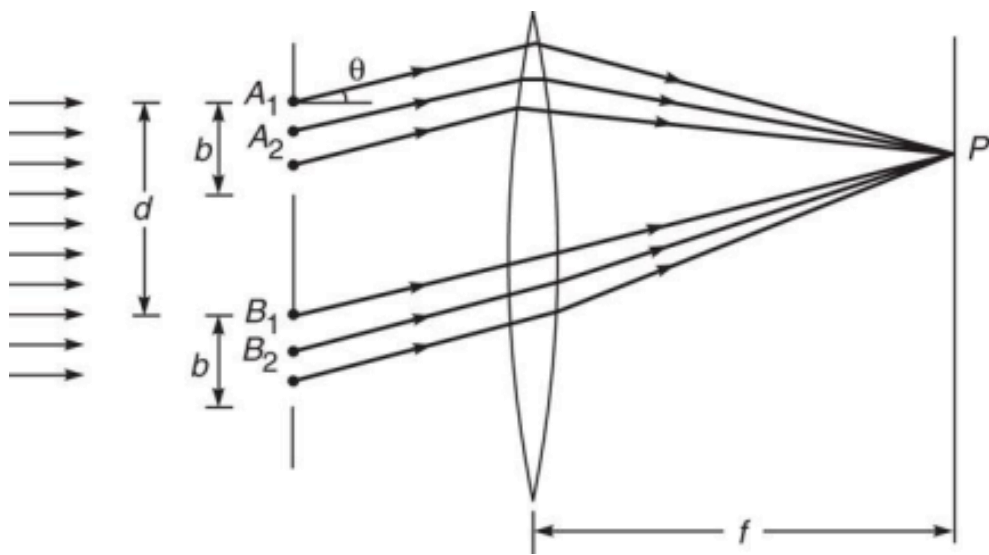
$$E = A \frac{\sin \beta}{\beta} \times 2 \cos\left(\omega t - \beta - \frac{\Phi_1}{2}\right) \cos\left(\frac{\Phi_1}{2}\right)$$

$$\therefore I = |E|^2 = 4 A^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma$$

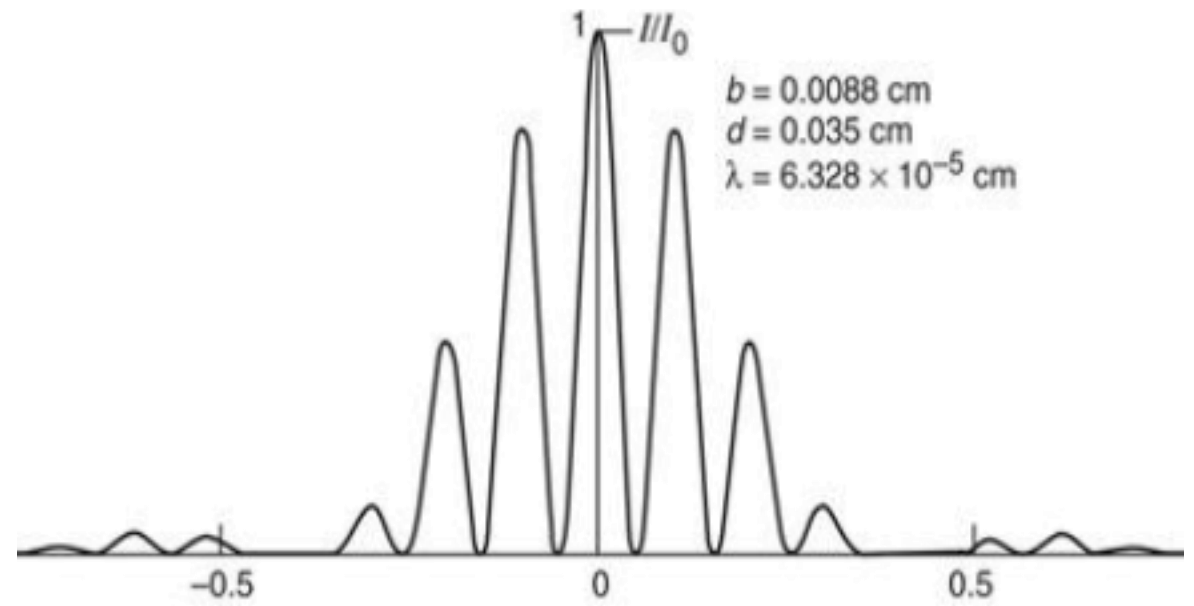
$$\text{Where } \gamma = \frac{\Phi_1}{2} = \frac{2\pi}{2\lambda} d \sin \theta = \pi d \sin \theta / \lambda$$

$$\text{Also } \beta = \frac{\pi}{\lambda} b \sin \theta$$

$$I = I_o \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma ; I_o = |2A|^2 \text{ is intensity at } \theta = 0$$

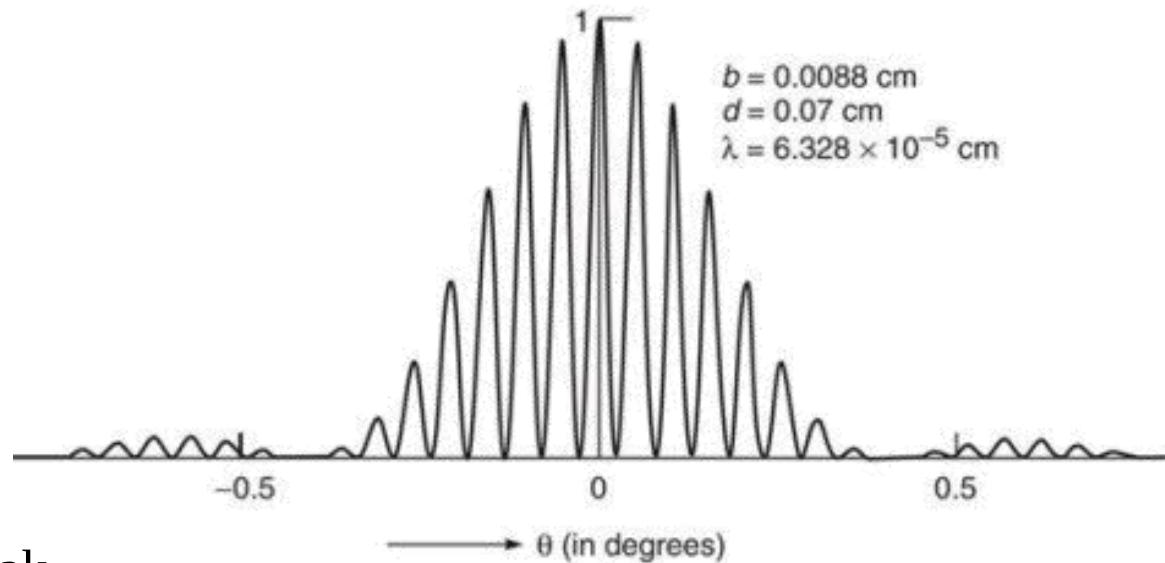


$$\frac{I}{I_o} = \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma$$



→ θ (in degrees)

Observe from the results:



Read: Text book Optics by Ajoy Ghatak

N-slit diffraction pattern (Grating):

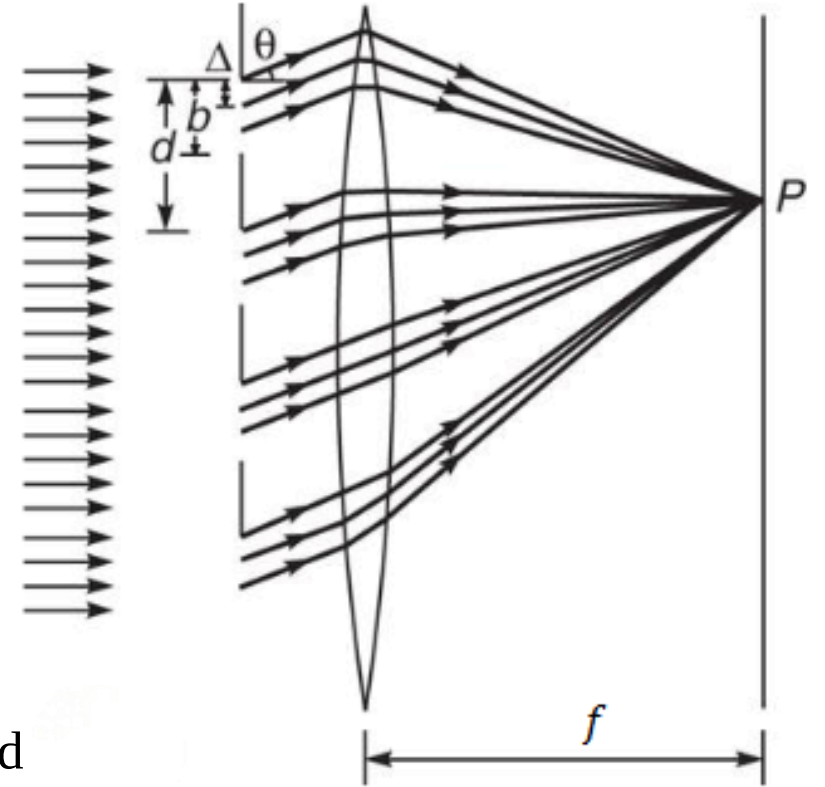
$$I = I_0 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$$

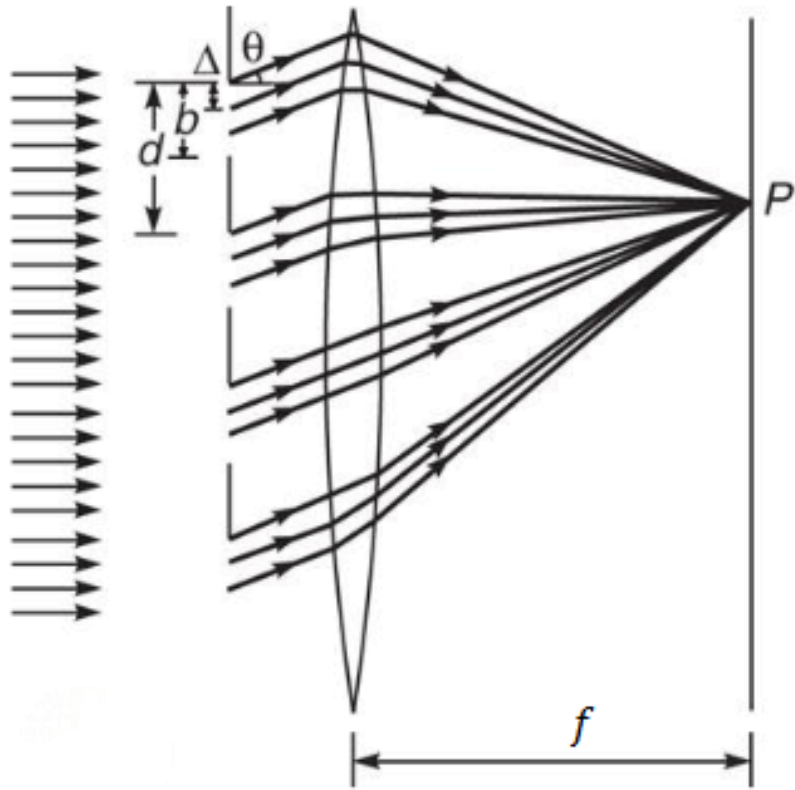
Where

$$\gamma = \frac{\pi}{\lambda} d \sin \theta; \quad \beta = \frac{\pi}{\lambda} b \sin \theta$$

$\frac{\sin^2 \beta}{\beta^2}$: Single slit diffraction

$\frac{\sin^2 N\gamma}{\sin^2 \gamma}$: Interference between the diffracted lights from N equally spaced point sources between neighboring slits





$$I = I_0 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$$

$$\gamma = \frac{\pi}{\lambda} d \sin \theta$$

For $N = 1$, we get single slit pattern $\because \left. \frac{\sin^2 N\gamma}{\sin^2 \gamma} \right|_{N=1} = 1$

For $N = 2$, we will get double slit pattern

$$\because \frac{\sin^2 2\gamma}{\sin^2 \gamma} = \frac{4 \sin^2 \gamma \cos^2 \gamma}{\sin^2 \gamma} = 4 \cos^2 \gamma,$$

$$\Rightarrow I = (2A)^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma$$

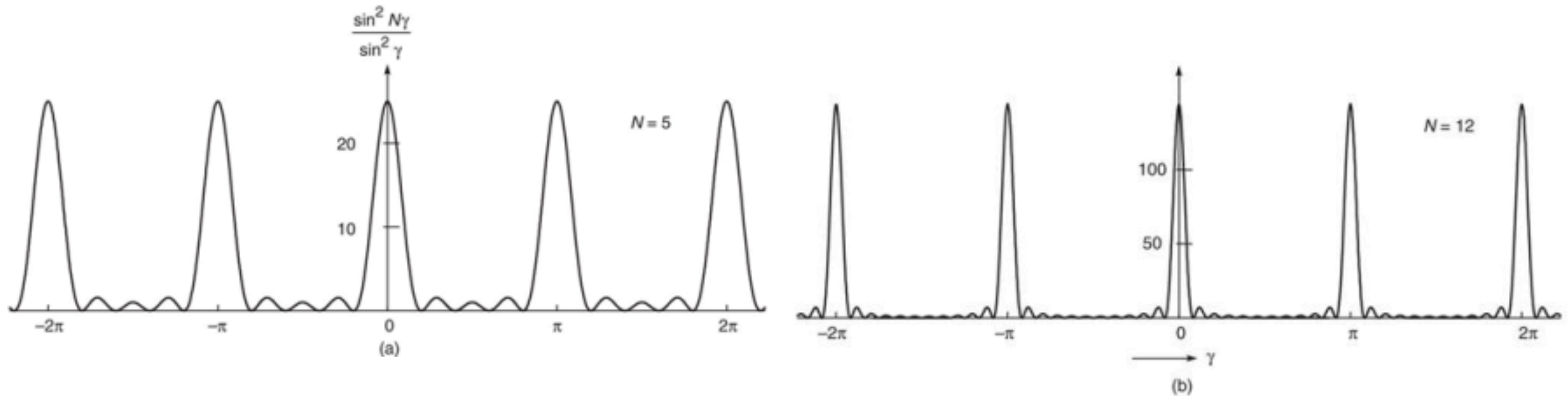
$$\Rightarrow I = I_0 \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma \quad \text{same as for 2 slits}$$

For large N,

$$\frac{\sin^2 N\gamma}{\sin^2 \gamma} \text{ peaks at } \gamma \approx m\pi \Rightarrow \frac{\pi d \sin\theta}{\lambda} = m\pi; m = 0, 1, 2, 3, \dots$$

$$\gamma = \frac{\pi}{\lambda} d \sin \theta$$

$$\Rightarrow d \sin \theta = m\lambda; \quad m = 0, 1, 2, 3, \dots : \text{Grating equation}$$



❖ Larger N shows sharper fringes (notice the difference in vertical scale)

Grating equation:

$$d \sin\theta = m\lambda; \quad m = 0, 1, 2, 3, \dots$$

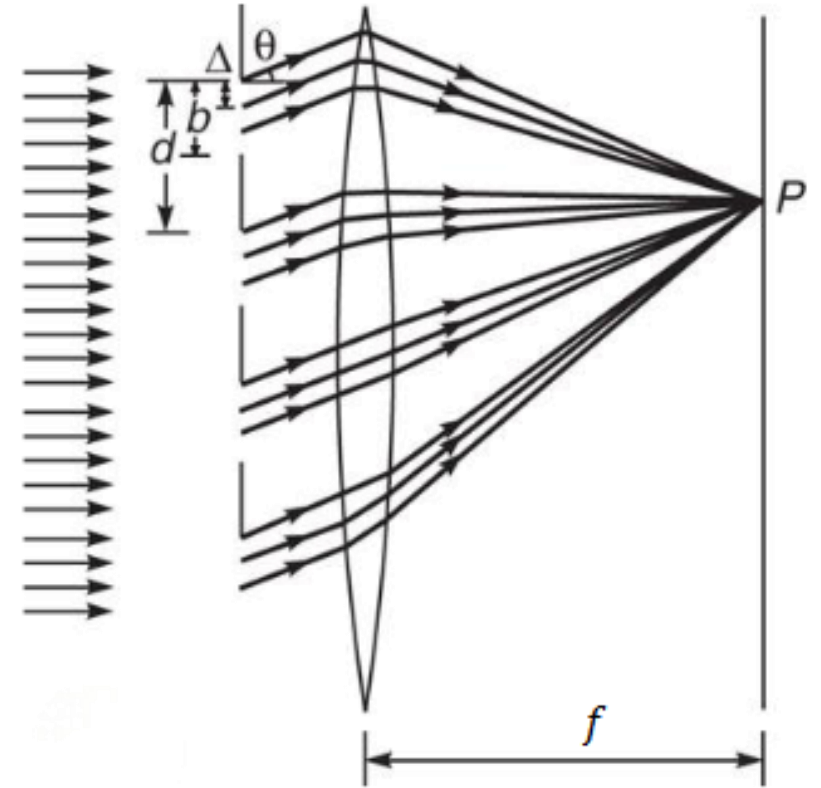
Where,

d is the grating element (distance between the neighbouring slits)

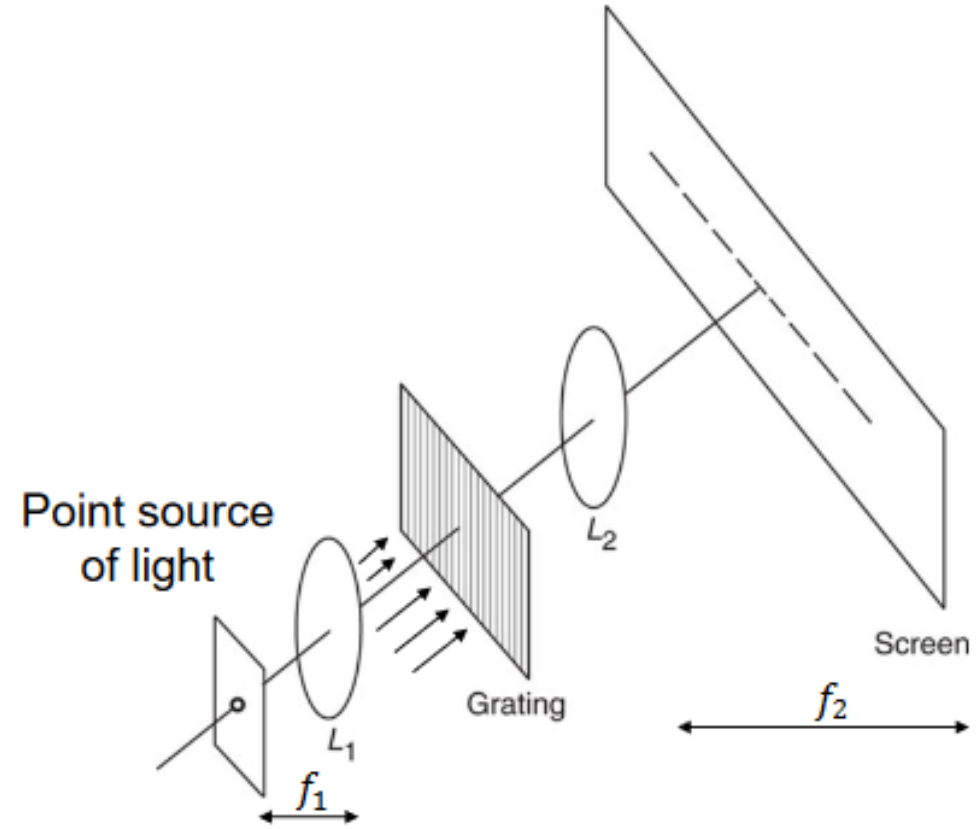
For $m \neq 0$

Each λ will diffract at different angles (θ 's) for a given d

$\Rightarrow$ Knowing m and measuring θ , one can determine λ for a given d



Schematic of a grating



Schematic of Fraunhofer diffraction of a grating

- In the original experiment of Rowland in 1827, there were 14438 grooves/inch, for which what will be value of grating element d ? 1 inch = 2.54 cm

$$d = \frac{2.54}{14438} = 1.759 \times 10^{-4} \text{ cm}$$

- For this grating, what will be the max order of the diffracted light of $\lambda = 6 \times 10^{-5} \text{ cm}$?

$$d \sin \theta = m\lambda$$

$$\Rightarrow m = \frac{d \sin \theta}{\lambda}$$

$$\Rightarrow m = \frac{1.759 \times 10^{-4}}{6 \times 10^{-5}} \sin \theta$$

$$\Rightarrow m \approx 2.93 \sin \theta \quad \Rightarrow m|_{\max} = 2$$

Resolving power of a grating is given by

$$R = \frac{\lambda}{\Delta\lambda} \approx mN$$

Where $\Delta\lambda$ is the wavelength separation between two neighbouring wavelengths.

- Thus if we wish to resolve the $D_{1,2}$ lines from the a sodium lamp, $\lambda = 5.89 \times 10^{-5}$ cm and $\Delta\lambda = 6 \text{ \AA}$.

$\Rightarrow$ For $m = 1$,

$$N = \frac{5.89 \times 10^{-5}}{6 \times 10^{-8}} \approx 1000$$

Thanks